

Quiz 11

Instructions: There are totally 2 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Use Laplace transform to solve the initial value problem:

$$x'' + 16x = 0; \quad x(0) = 3, \quad x'(0) = 2.$$

Let $X(s) = \mathcal{L}\{x(t)\}$, we have

$$\mathcal{L}\{x'(t)\} = s\mathcal{L}\{x(t)\} - x(0) = sX(s) - 3$$

$$\mathcal{L}\{x''(t)\} = s^2\mathcal{L}\{x(t)\} - sx(0) - x'(0) = s^2X(s) - 3s - 2$$

Then

$$\mathcal{L}\{x'' + 16x\} = \mathcal{L}\{0\}.$$

$$\mathcal{L}\{x''\} + 16\mathcal{L}\{x(t)\} = 0$$

$$(s^2X(s) - 3s - 2) + 16X(s) = 0$$

$$(s^2 + 16)X(s) = 3s + 2$$

$$X(s) = \frac{3s + 2}{s^2 + 16}$$

Thus

$$x(t) = \mathcal{L}^{-1}\{X(s)\}$$

$$= \mathcal{L}^{-1}\left\{\frac{3s + 2}{s^2 + 16}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{3s}{s^2 + 16}\right\} + \mathcal{L}^{-1}\left\{\frac{2}{s^2 + 16}\right\}$$

$\hookrightarrow 4^2 \qquad \qquad \qquad \hookrightarrow 4^2$

$$= 3 \cos 4t + \frac{1}{2} \sin 4t$$

Problem 2. (4 points) Find the inverse Laplace transform of $\frac{1}{s(s^2 + 4)}$ using the formula

$$\mathcal{L}^{-1}\left(\frac{F(s)}{s}\right) = \int_0^t f(\tau) d\tau \text{ where } f(t) = \mathcal{L}^{-1}\{F(s)\}.$$

$$\text{Take } F(s) = \frac{1}{s^2 + 4} \Rightarrow \frac{F(s)}{s} = \frac{1}{s(s^2 + 4)}$$

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s^2 + 4}\right\} = \frac{1}{2} \sin 2t$$

Thus

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2 + 4)}\right\} = \int_0^t \frac{1}{2} \sin 2\tau d\tau = \left[-\frac{\cos 2\tau}{4}\right]_{\tau=0}^{\tau=t}$$

$$= -\frac{1}{4}(\cos 2t - 1)$$